CHAIN-WELL MACHINERY.—There will be fitted and attached to the sides of each man-hole, on a line plumb with the lock side of the well, cheek timbers of white oak, which will be bolted to the walls by rag-bolts $\frac{7}{8}$ of an inch diameter. In the opening well, the timbers will be framed to suit the rake of the man-hole, and one side will have a vertical drum-pulley fitted to a footstep prepared to receive the bottom of the spindle, and the top will be fitted by a special form of brass to a bracket bolted to the timber by wood screws. The cast iron cheek-plates will be the same for all the wells, and will be fitted and bolted to the timbers by wood-screws. Large sheave-pulleys of cast iron, having spindles 2 inches diameter and journals 3 inches long, must be provided. The spindle to be keyed fast to sheave. The journals will have brass boxes made to fit into ways cast in the cheek-plates for them. These boxes will be kept in place by cast iron blocks made to fit into ways prepared on the cheek-plates. The blocks are kept in place by loosely-fitting wrought iron keys. This machinery will require to be put in place and fitted to the timber with great care, as the sheaves and pulleys must work perfectly true when completed.

CRABS.—For working the gates to be supplied to each well, or eight to the lock. They will be formed for single and double purchase, and will be made up so that only a part of the gearing will be attached to frame. The drum and bevel spur-wheel will be fitted to the coping into which checks must be cut on each side of well-opening for the spindle boxes. These checks will be 6 inches broad, 3 inches wide and $8\frac{1}{4}$ inches deep, measured on centre line. The boxes are to be of cast iron, bushed with brass and bored out for spindle as shown ; oil-ways and cellars to be formed in casting.

The drums and bevel wheel to be cast in one to the dimensions and form shown on drawing ; cored and finished for spindle, which must be of wrought iron $2\frac{1}{2}$ inches by $2\frac{1}{2}$ inches, turned up at the ends $3\frac{5}{8}$ inches long for journal. The spindle to be a loose fit through drum. The frame is to be a dome-shaped casting, pierced on sides and top as shown ; it must have a base moulding, with projections at three points for feet. On these fitting-strips must be cast and be truly turned up on bottom so as to be at right angles to the spindles. As before stated, the crabs are to have double purchase, and for this purpose two spindles are provided having sockets on the top carefully fitted into recesses prepared in frame. Bevel pinion and spur-wheel, 18 inches diameter, is to be fitted to first power spindle, the foot of which will work in a step-plate bushed with brass and bolted to snugs cast on frame as shown. The second power spindle will carry a pinion 6 inches diameter, geared into spur-wheel. This couple to have $2\frac{1}{2}$ inch face and 1 inch pitch. The bevel wheels to have 3 inches face and $1\frac{1}{2}$ inch pitch. All the teeth will have to be machine cut, and must be fitted to work without *back lash*. The frame will be bolted to the coping by fox-tail bolts let 5 inches into the stone. Bolts must be fitted into the rim of the large drum for fastening the chain to the same.

CAPSTAN.—A wrought iron capstan of such dimensions and form as may be directed must be provided for each crab.

DRAG CHAIN—To be of the best class of thoroughly tested $\frac{9}{16}$ or $\frac{5}{8}$ inch chain or such other weight as may be directed—one end to be attached to the drum of the crab by an eye-bolt, and at the other end there is to be a link made $\frac{7}{8}$ of an inch round iron to connect with drag-hook.

WROUGHT IRON.—The tie-rods, suspension gear and anchor-rods must be made from double-refined wrought iron, the ultimate strength of which, in tension, must not be less than 50,000 pounds per square inch, or the limit of elasticity less than 25,000 pounds per square inch. The ductility of the iron must be equal to an extension of 20 per cent. before rupture.